

Why Most Crypto Backtests Lie

12 Failure Modes from Building an Audit-First Crypto Research Lab

Vertico Research

Educational guide. This publication uses sanitized portfolio evidence. It does not disclose restricted data, private logs, machine paths, unpublished source code, or sensitive implementation details.

1. **Survivorship bias.** Reconstruct the universe as it was known at each decision time.
2. **Point-in-time universe errors.** Later listings, delistings, and revisions cannot enter earlier decisions.
3. **Same-bar execution.** A signal cannot use information that was unavailable at the execution price.
4. **Forward scanning for missing fills.** Future observations cannot repair an unavailable execution price.
5. **Gap and quarantine leakage.** Use a frozen exclusion rule for unavailable or quarantined records.
6. **Fold contamination.** Keep tuning and transformations inside their permitted partition.
7. **Underestimated fees and slippage.** Missing executable prices and unfrozen costs limit interpretation.
8. **Lucky parameters and random seeds.** Record the search space and distinguish selection from confirmation.
9. **Confusing test-suite success with profitability.** Tests validate software contracts, not an economic edge.
10. **Ignoring bootstrap uncertainty.** Point estimates do not communicate sampling variation.
11. **Ignoring DSR and multiple testing.** More trials raise false-positive risk.
12. **Opening the final holdout too early.** A holdout seen during development is no longer independent.

Public evidence boundary

The sanitized public snapshot records zero capital permission, no accepted strategy P&L, and sealed final holdouts. Source checkpoint 5303653, the requested v5 cleanroom evaluation, and AUDIT_PENDING_SERVICE_RECOVERY are unavailable through the local publication adapter. No unavailable performance figures are published.

Disclosure

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